

ECE321

Electronic circuits

Lecture (03)- Frequency Response_02

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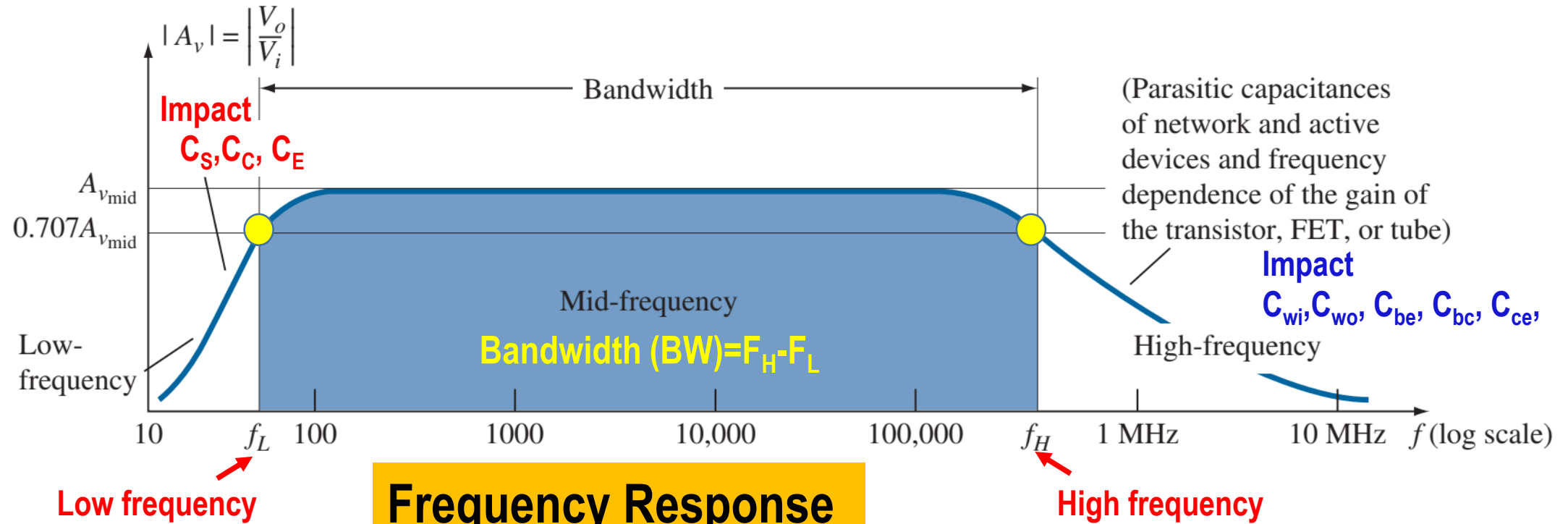
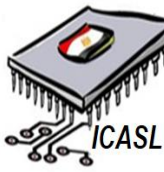


- ☐ Typical Frequency Response
- ☐ Low-Frequency response —BJT
- ☐ Exercise
- ☐ High-Frequency Response —BJT
- ☐ Miller Theory
- ☐ Exercise
- ☐ Notes



- ☐ **Typical Frequency Response**
- ☐ **Low-Frequency response —BJT**
- ☐ **Exercise**
- ☐ **High-Frequency Response —BJT**
- ☐ **Miller Theory**
- ☐ **Exercise**
- ☐ **Notes**

Typical Frequency Response



Frequency Response

- F_L and F_H :
- Corner Frequency
 - Cut off frequency
 - Break frequency
 - Half-power frequency
 - -3dB Frequency

$$P_{o,mid} = \frac{V_{o,mid}^2}{R_o}$$

$$P_{o,corner} = \frac{V_{o,corner}^2}{R_o}$$

$$= \frac{V_{o,mid}^2}{2 * R_o}$$

Where, $V_{o,corner} = \frac{V_{o,mid}}{\sqrt{2}}$

$P_{o,corner} = \frac{P_{o,mid}}{2}$





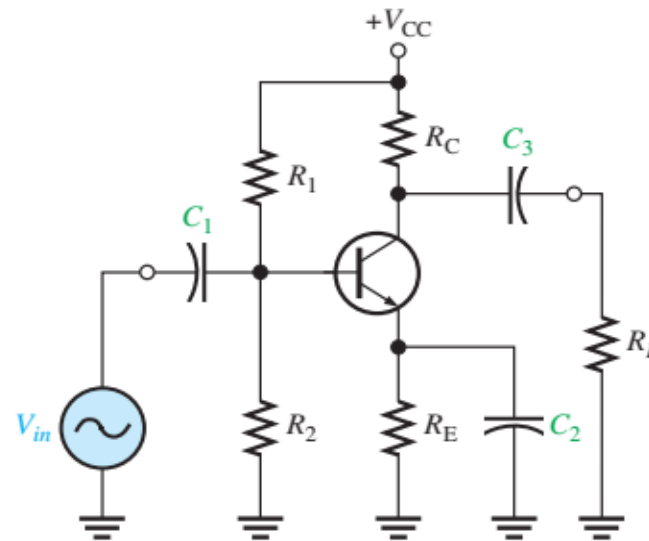
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Low-Frequency Response —BJT (1/6)

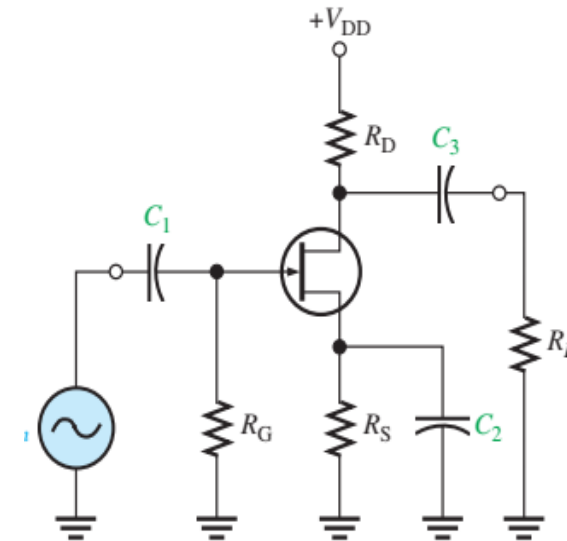


❑ Effect of Coupling Capacitors

- ❑ **At lower frequencies** the reactance is greater, and it decreases as the frequency increases, $X_C = \frac{1}{2\pi fC}$
- ❑ **At lower frequencies**, audio frequencies below 10 Hz—capacitively coupled amplifiers have less voltage gain than they have at higher frequencies.
 - ❑ The reason is that at **lower frequencies** more signal voltage is dropped across C_1 and C_3 because their reactances are higher.
 - ❑ This higher signal voltage drop at lower frequencies reduces the voltage gain.



BJT



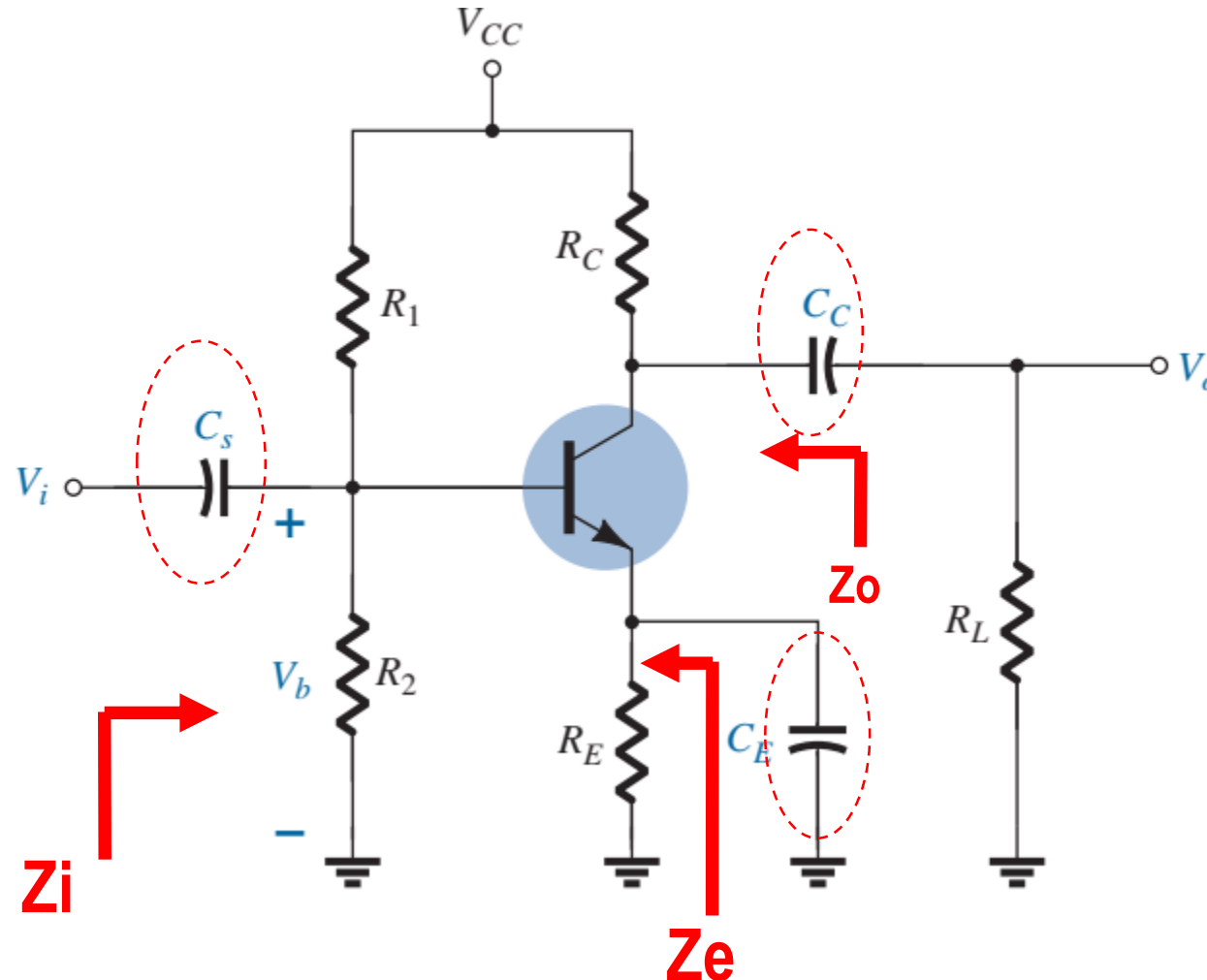
FET

Low-Frequency Response —BJT (2/6)

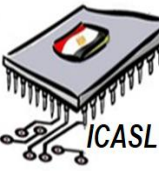


- ❑ The capacitors C_S , C_C , and C_E will determine the **low-frequency response**.
- ❑ We will now examine the impact of each **independently** 😊

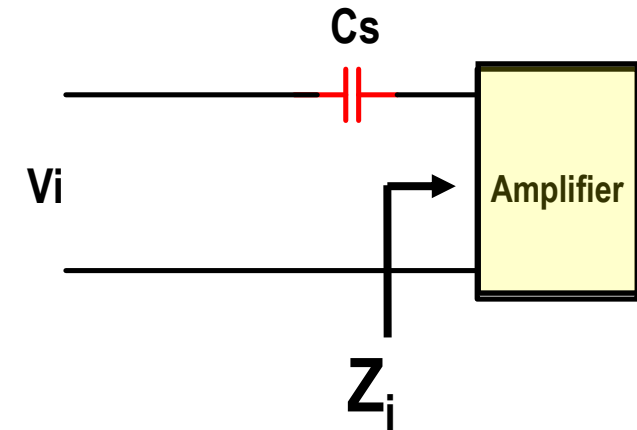
1. Effect of C_S
2. Effect of C_C
3. Effect of C_E



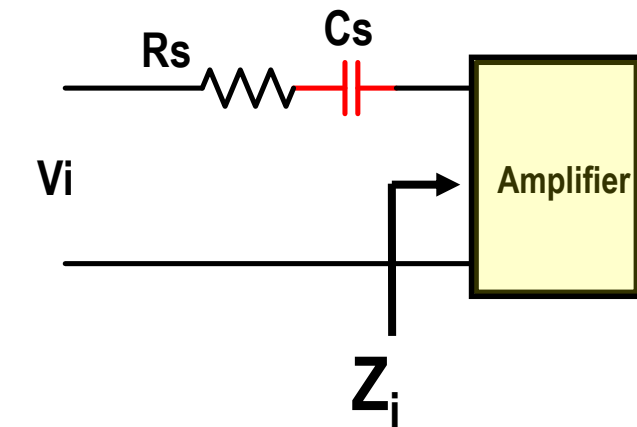
Low-Frequency Response —BJT (3/6)



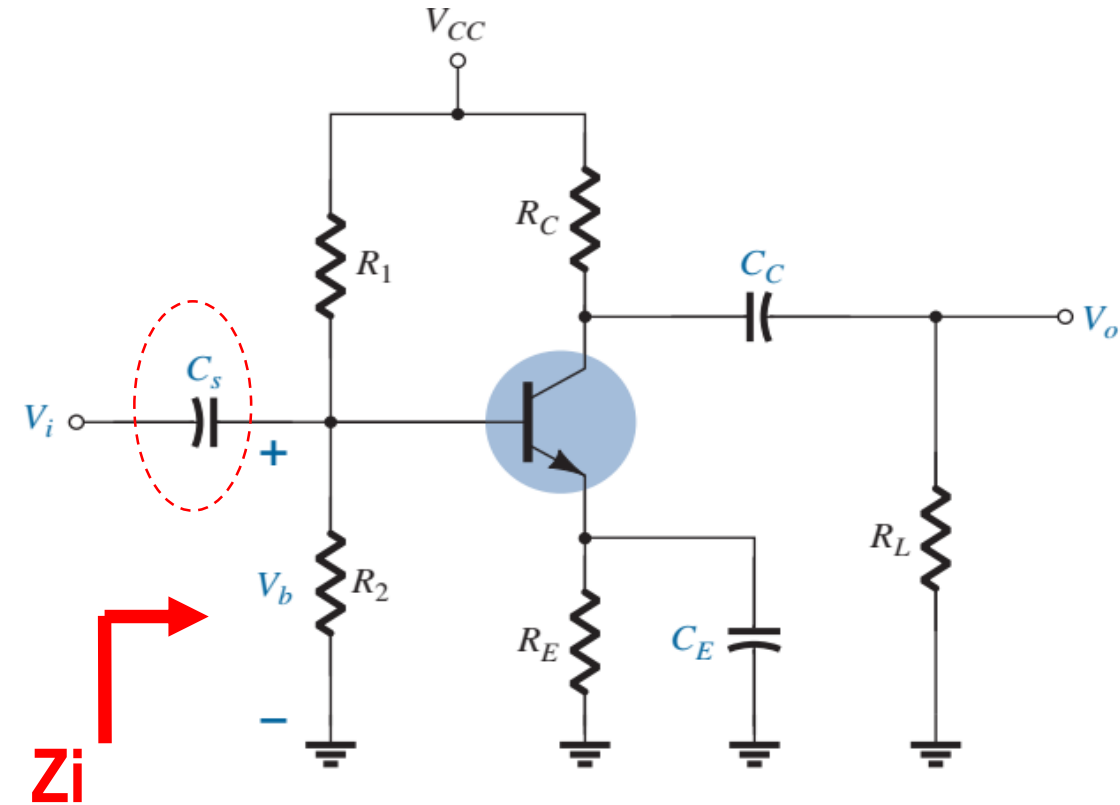
1. Effect of C_s



$$F_{LS} = \frac{1}{2\pi(z_i)C_s}$$



$$F_{LS} = \frac{1}{2\pi(Z_i + R_s)C_s}$$



Note:

Z_i is calculated from AC analysis

$Z_i = R_1 || R_2 || (\beta * r_e)$ for the shown circuit.

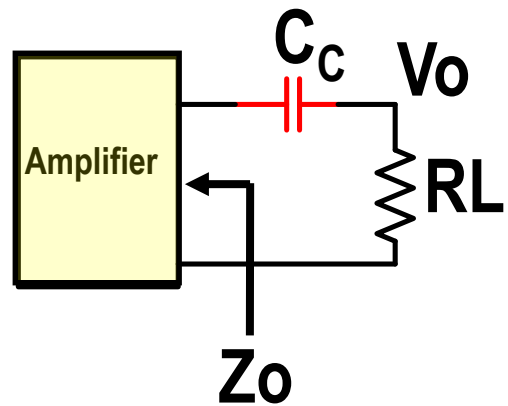
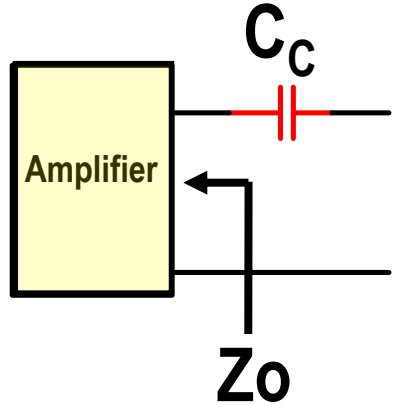
For Each circuit, you have to do DC and AC analysis



Low-Frequency Response —BJT (4/6)

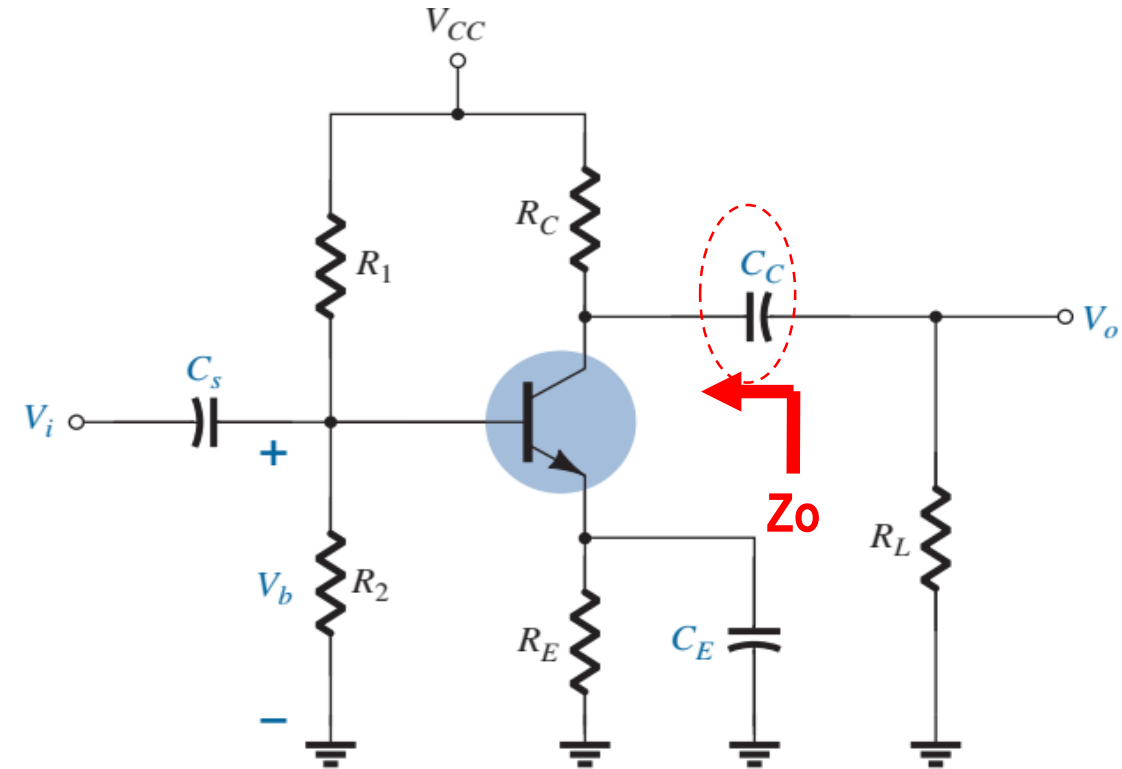


2. Effect of C_C



$$F_{LC} = \frac{1}{2\pi(Z_o)C_C}$$

$$F_{LC} = \frac{1}{2\pi(Z_o + R_L)C_C}$$



Note:

Z_o is calculated from AC analysis

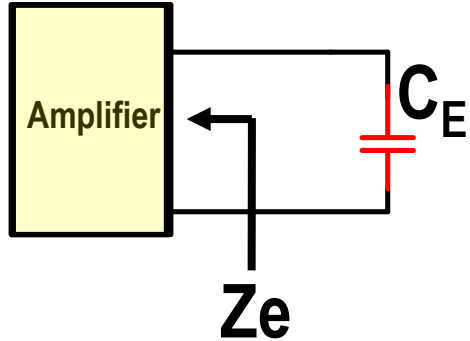
$Z_o = R_C || R_o$ for the shown circuit

For each circuit, you have to do DC and AC analysis

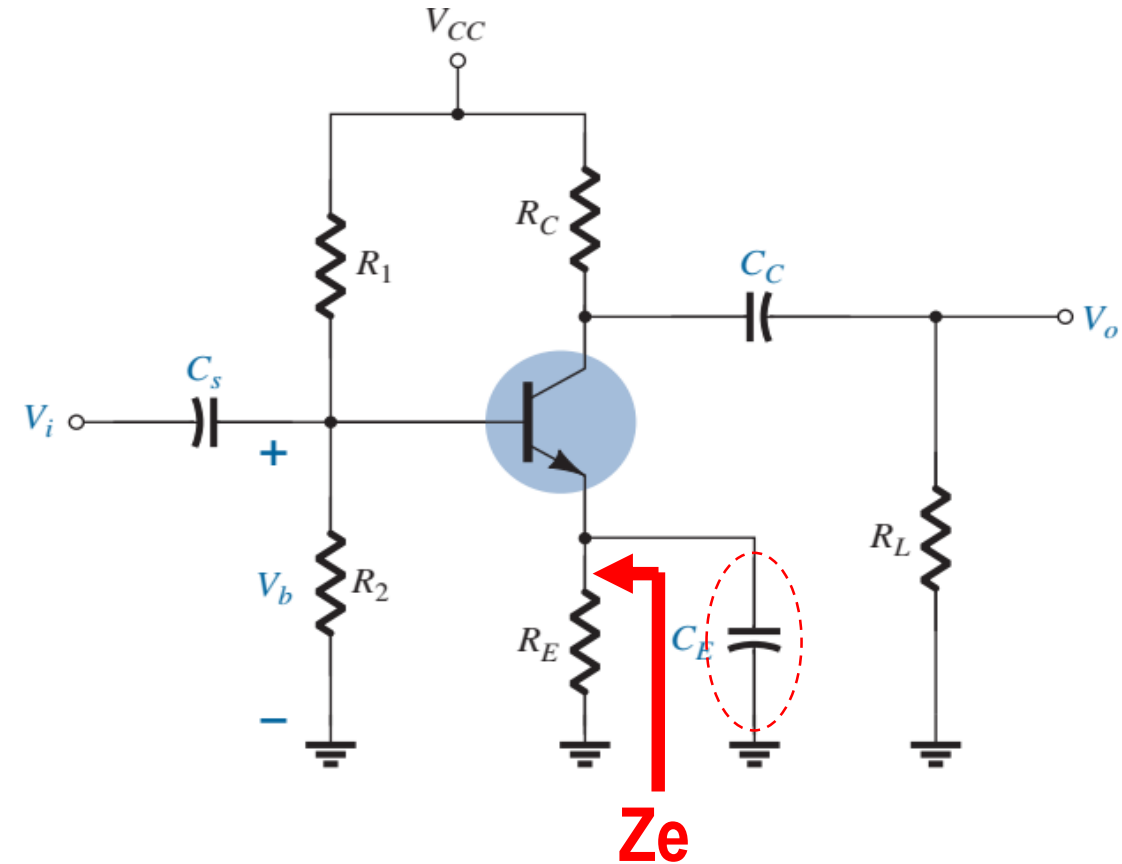
Low-Frequency Response —BJT (5/6)



3- Effect of C_E



$$F_{LE} = \frac{1}{2\pi(Z_e)C_E}$$



Note:

Z_e is calculated from AC analysis

$Z_e = R_E \parallel [\{ (R_1 \parallel R_2 \parallel R_s) + \beta * r_e \} / \beta]$ for the shown circuit

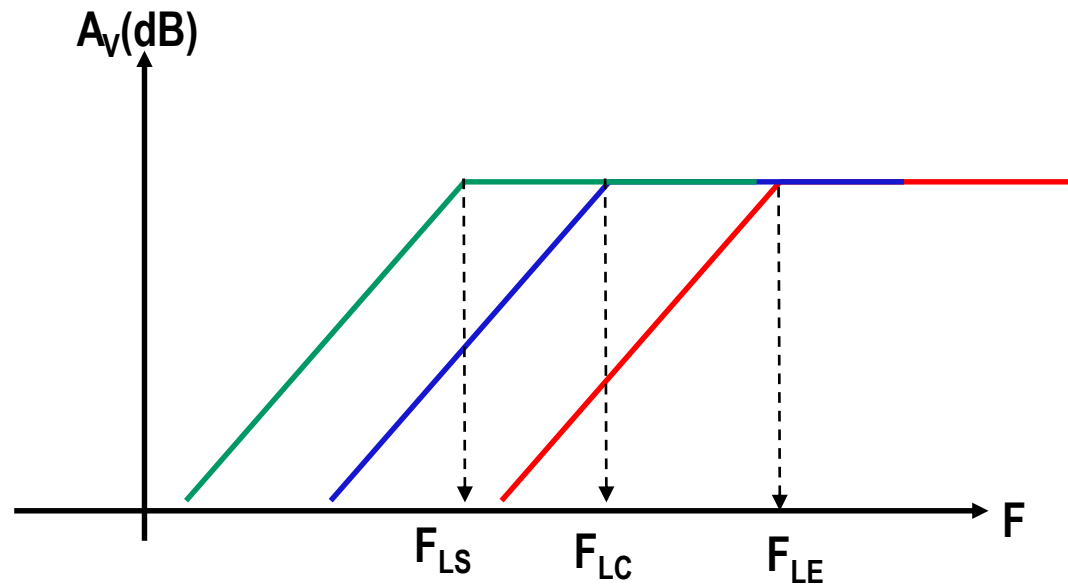
For Each circuit, you have to do AC analysis

Low-Frequency Response —BJT (6/6)

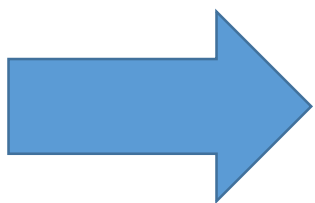


Notes

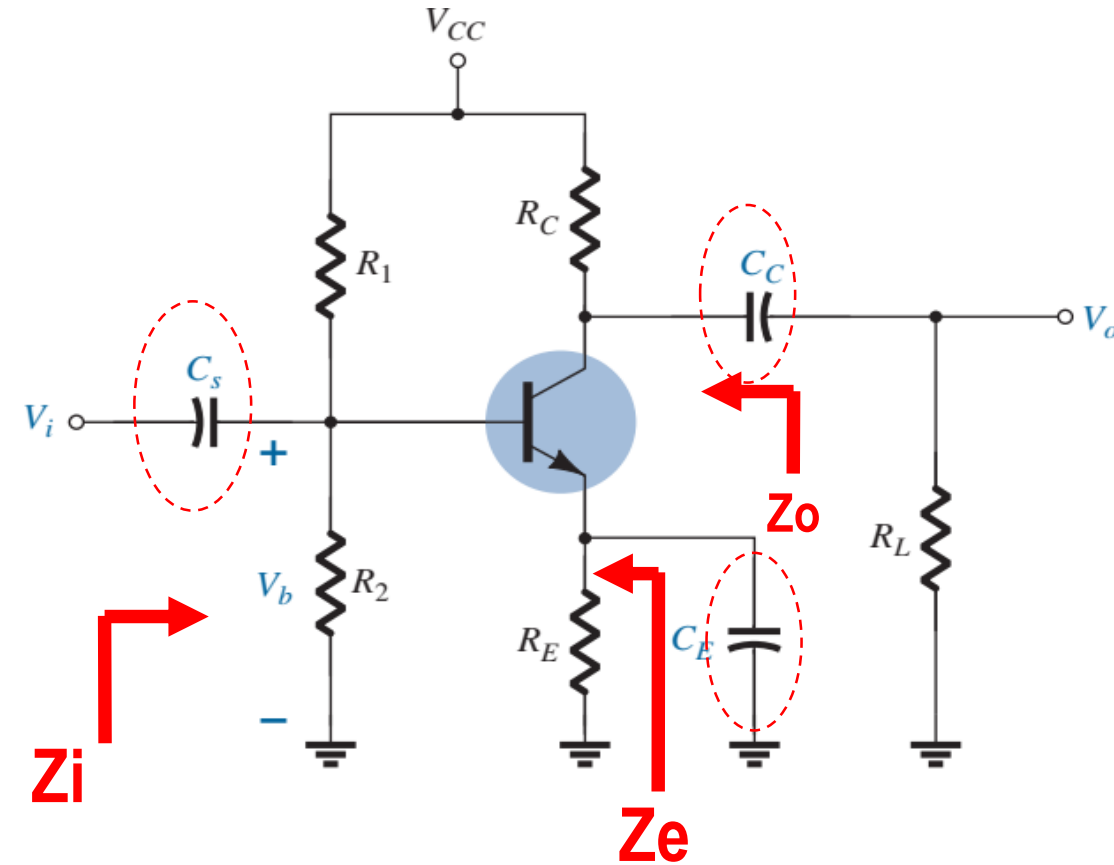
- There are three corners F_{LS} , F_{LC} , F_{LE}
- The question is which one is the **dominated corner frequency** for the amplifier



$$\begin{aligned} F_{LE} &= 10\text{KHz} \\ F_{LC} &= 1\text{KHz} \\ F_{LS} &= 10\text{Hz} \end{aligned}$$



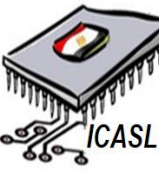
$$F_L = F_{LE}$$





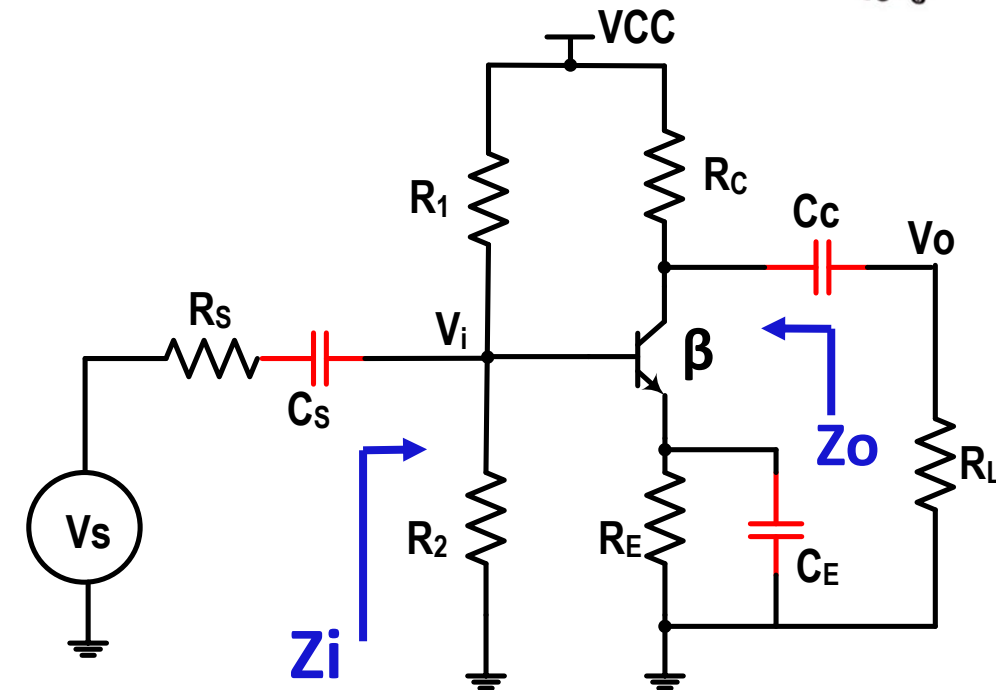
- ☐ Typical Frequency Response
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- ☐ **Exercise**
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Exercise (1/6)



□ For the shown amplifier, determine

1. r_e
2. Z_i , and Z_o
3. $A_v(\text{mid}) = \frac{V_o}{V_i}$, and $A_{vs}(\text{mid}) = \frac{V_o}{V_s}$
4. F_{LS} , F_{LC} , F_{LE}
5. F_L
6. Sketch the asymptote of the bode plot defined the cut-off frequencies in (4)
7. Sketch the low frequency response of the amplifier using the results of (5)



$$V_{CC}=14 \text{ V} , \beta=120$$

$$R_S=0.86\text{k}\Omega , R_1= 68\text{K}\Omega, R_2=10\text{k}\Omega$$

$$R_C=5.5\text{K}\Omega , R_E=1.2\text{k}\Omega, R_L=3.3\text{k}\Omega$$

$$C_S=0.47\mu\text{F}, C_C=0.47\mu\text{F}, C_E=20\mu\text{F}$$

Exercise (2/6)



1-To get r_e , we need to do DC analysis since $r_e = \frac{26mV}{I_E}$

□ DC Analysis

All capacitor $\rightarrow$ O.S

Assume BJT in active region

($V_{BE}=0.7$, $V_{CE} \geq 0.2$ V)

For Voltage divider , we can use exact solution

$$R_{TH} = R_1 || R_2 = 8.7k\Omega$$

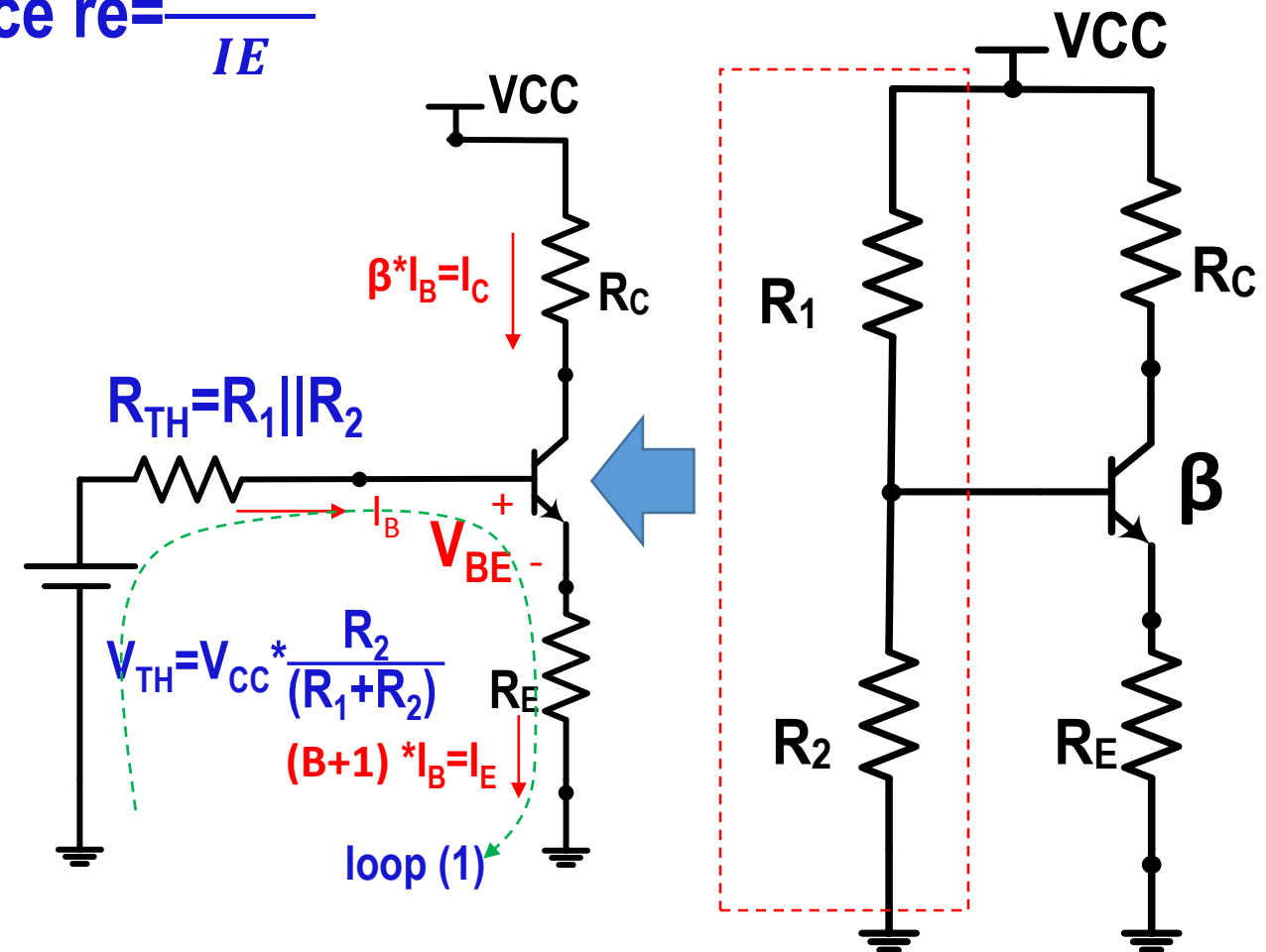
$$V_{TH} = V_{CC} * \frac{R_2}{(R_1 + R_2)} = 1.8 \text{ V}$$

To get I_B By using loop (1)

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1)R_E} = 7.1 \mu A$$

$$I_E = (\beta + 1) * I_B = 0.86mA \cong 1mA$$

Check
 $V_{CE} = V_C - V_E \geq 0.2$
✓



$\Rightarrow r_e = \frac{26mV}{I_E} = \frac{26mV}{1m} = 26 \Omega$

Exercise (3/6)

2-To get Z_i and Z_o , we need to do AC analysis

□ AC analysis

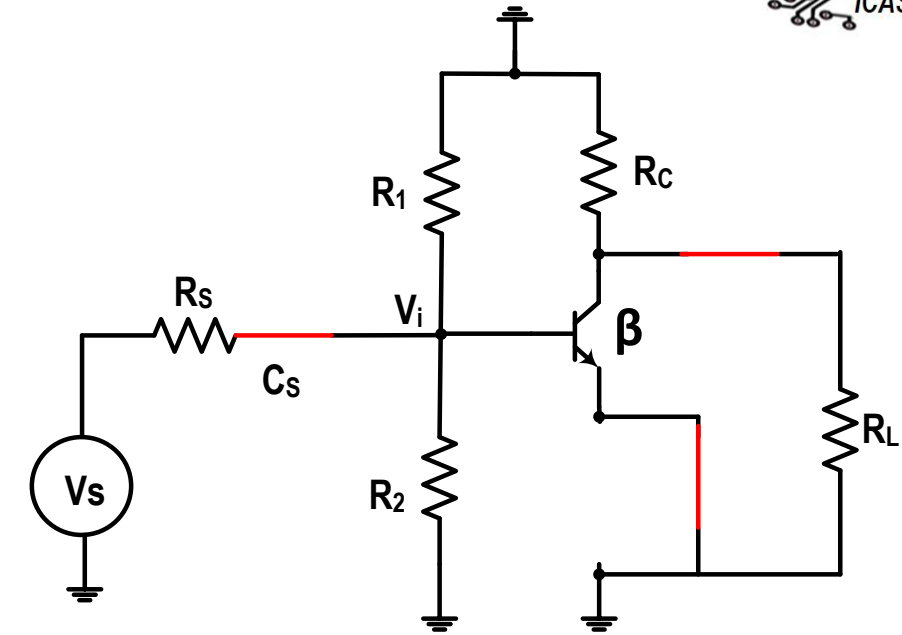
All DC voltage sources $\rightarrow$ S.C (GND)

All capacitor $\rightarrow$ S.C

$$Z_i = R_1 || R_2 || (\beta * r_e) = 2.29K\Omega$$

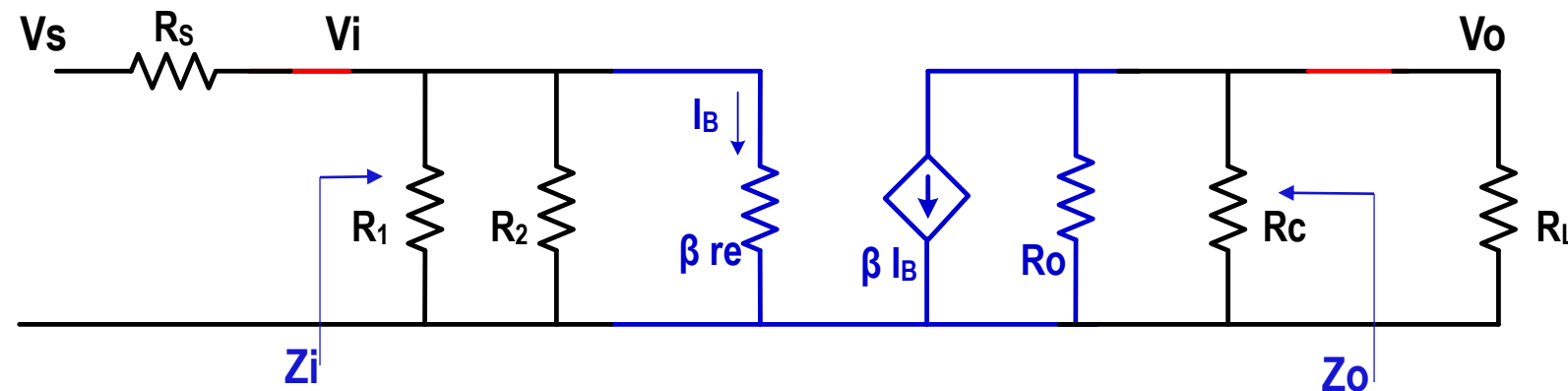
$$Z_o = R_o || R_c = R_c = 5.5K\Omega$$

3-To get $A_{v(mid)}$ and $A_{vs(mid)}$, we need to do AC analysis



$$A_{v(mid)} = \frac{V_o}{V_i} = \frac{-\beta I_B * (R_o || R_C || R_L)}{I_B * \beta * r_e}$$

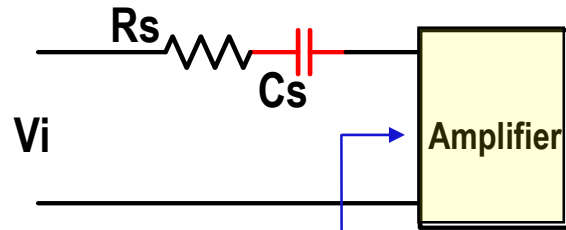
$$= \frac{-(R_C || R_L)}{r_e} = -79.3$$



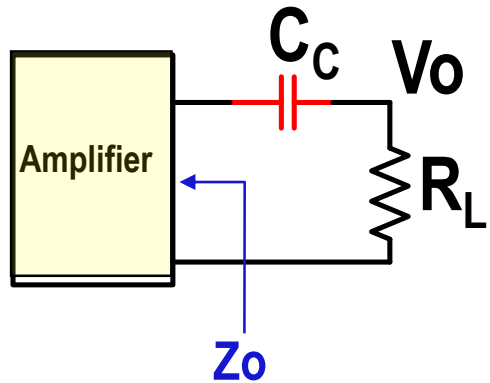
$$A_{vs(mid)} = \frac{V_o}{V_s} = \frac{V_o}{V_s} * \frac{V_i}{V_i} = \frac{V_o}{V_i} * \frac{V_i}{V_s} = A_{v(mid)} * \frac{V_i}{V_s} = A_{v(mid)} * \frac{Z_i}{Z_i + R_s} = -57.64$$

Exercise (4/6)

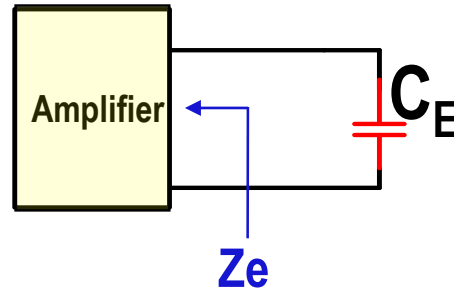
4-To get F_{LS} , F_{IC} , and F_{LE}



$$F_{LS} = \frac{1}{2\pi(Z_i + R_s)C_s} = 107.5 \text{ Hz}$$



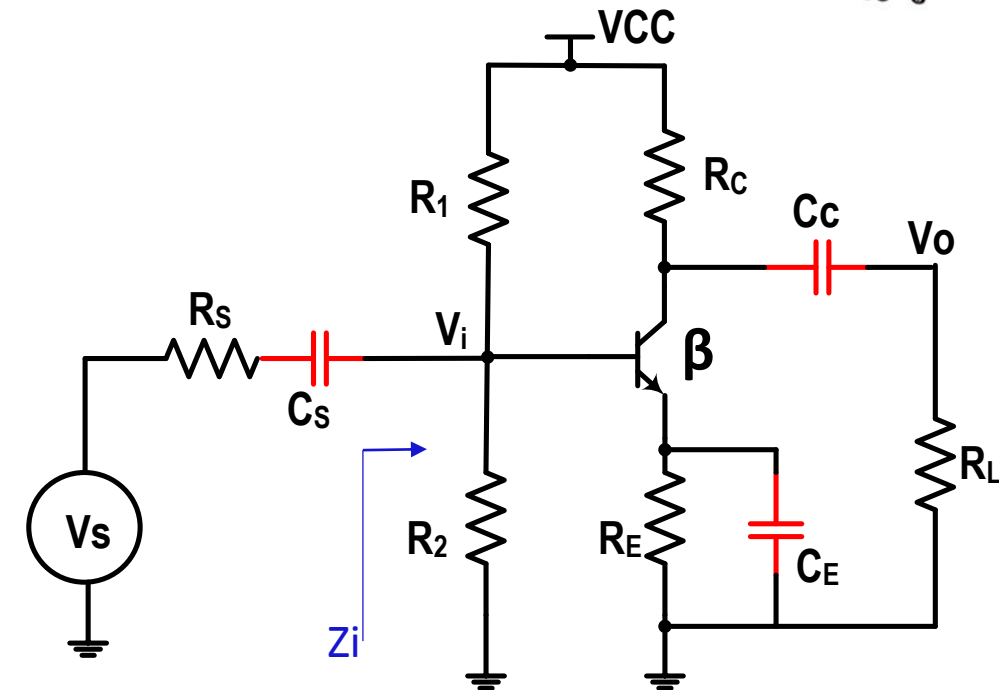
$$F_{LC} = \frac{1}{2\pi(Z_o + R_L)C_c} = 48.4 \text{ Hz}$$



$$F_{LE} = \frac{1}{2\pi(Z_e)C_E}$$

$$Z_e = R_E \parallel \frac{(R_1 \parallel R_2 \parallel R_s) + \beta r_e}{\beta} \cong 31.6 \Omega$$

$$F_{LE} = \frac{1}{2\pi(Z_e)C_E} = 251.3 \text{ Hz}$$



5-To get F_L

$$F_{LS} = 107.5 \text{ Hz}$$

$$F_{LC} = 48.4 \text{ Hz}$$

$$F_{LE} = 251.3 \text{ Hz}$$



$$F_L = F_{LE} = 251.3 \text{ Hz}$$

Exercise (5/6)

6- to draw the asymptotes of bode plot

$$F_{LS} = 107.5 \text{ Hz}$$

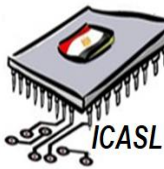
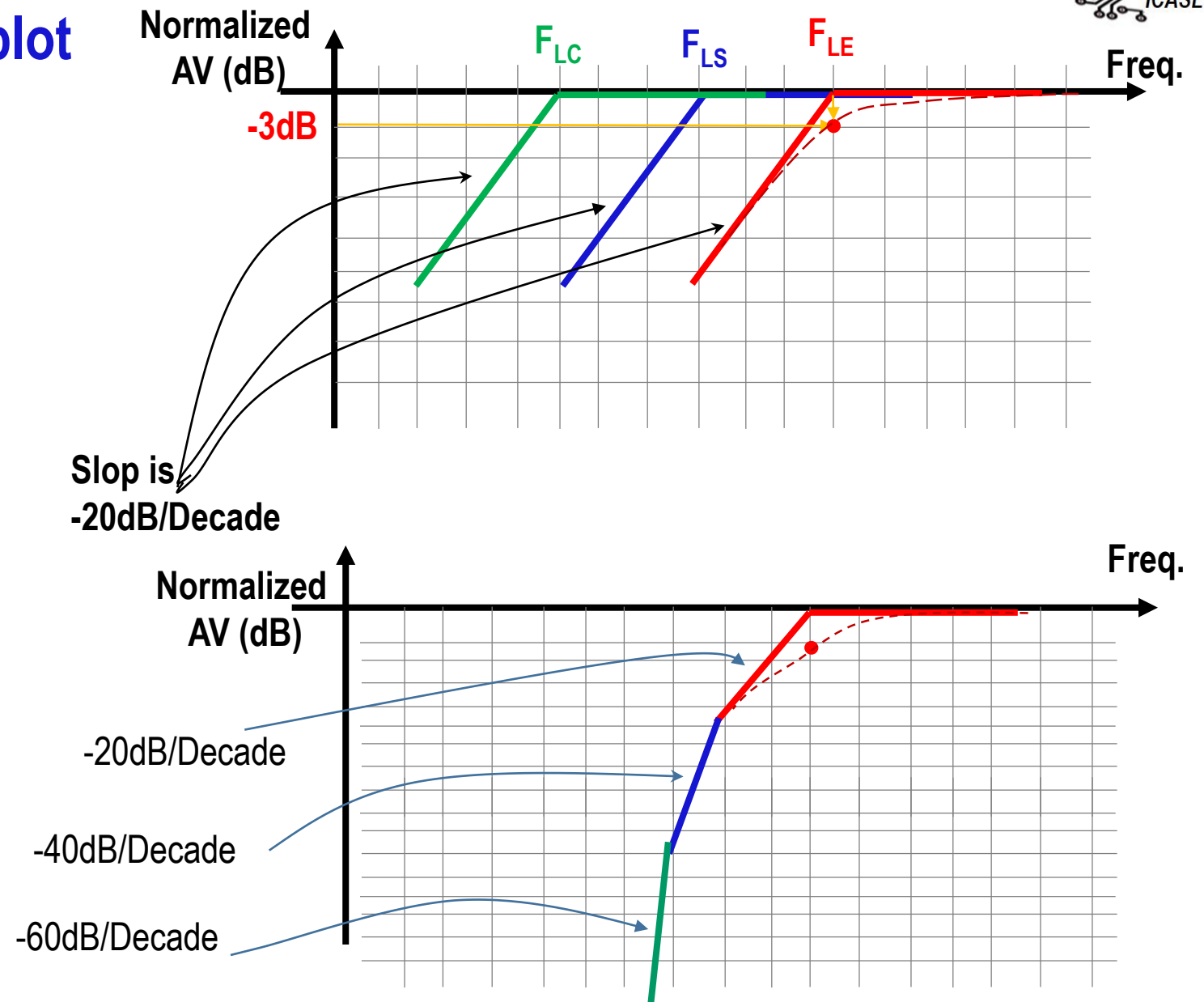
$$F_{LC} = 48.4 \text{ Hz}$$

$$F_{LE} = 251.3 \text{ Hz}$$

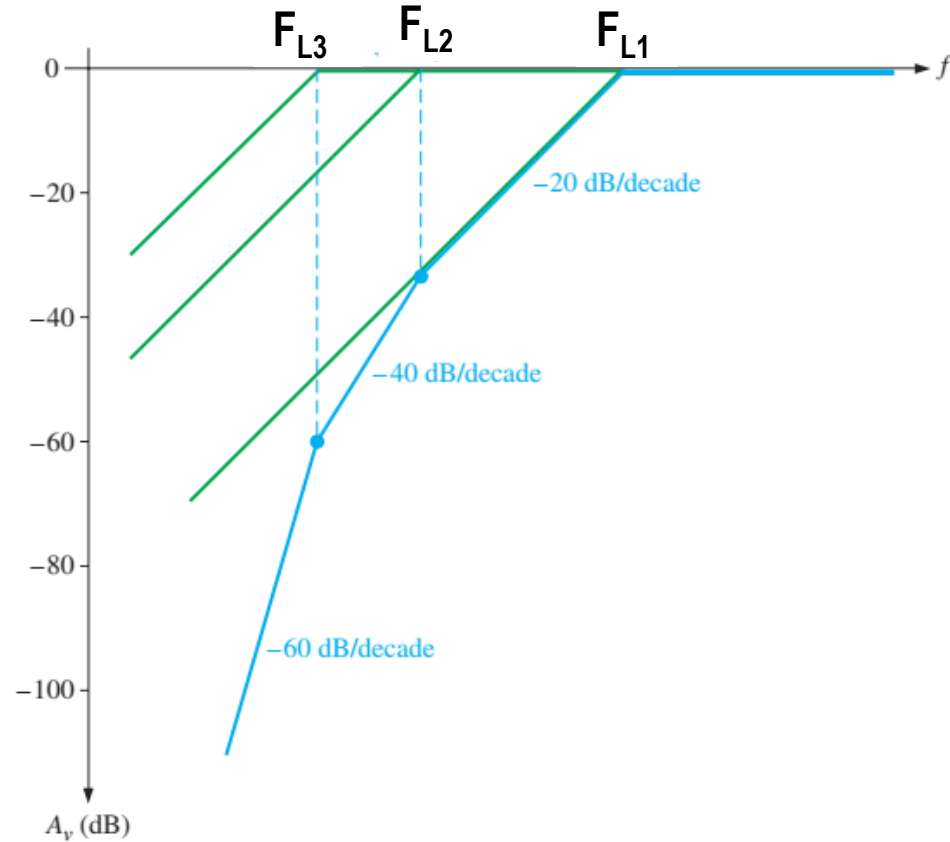


$$F_L = F_{LE} = 251.3 \text{ Hz}$$

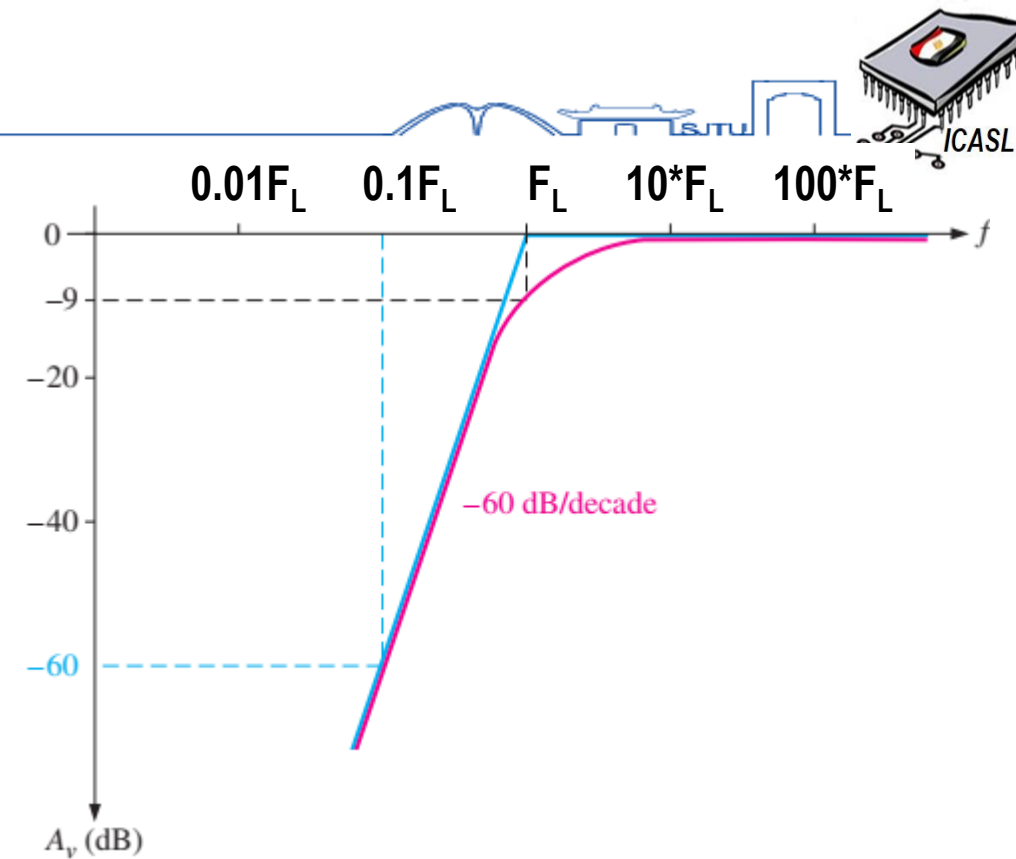
7-To Sketch the low frequency response of the amplifier using the results of (5)



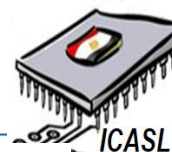
Exercise (6/6)



Composite Bode plot of a BJT amplifier response for **three low-frequency RC** circuits with **different critical frequencies**. Total response is shown by the blue curve.



Composite Bode plot of an amplifier response where all RC circuits have the **same F_L** . (Blue is ideal; red is actual.)





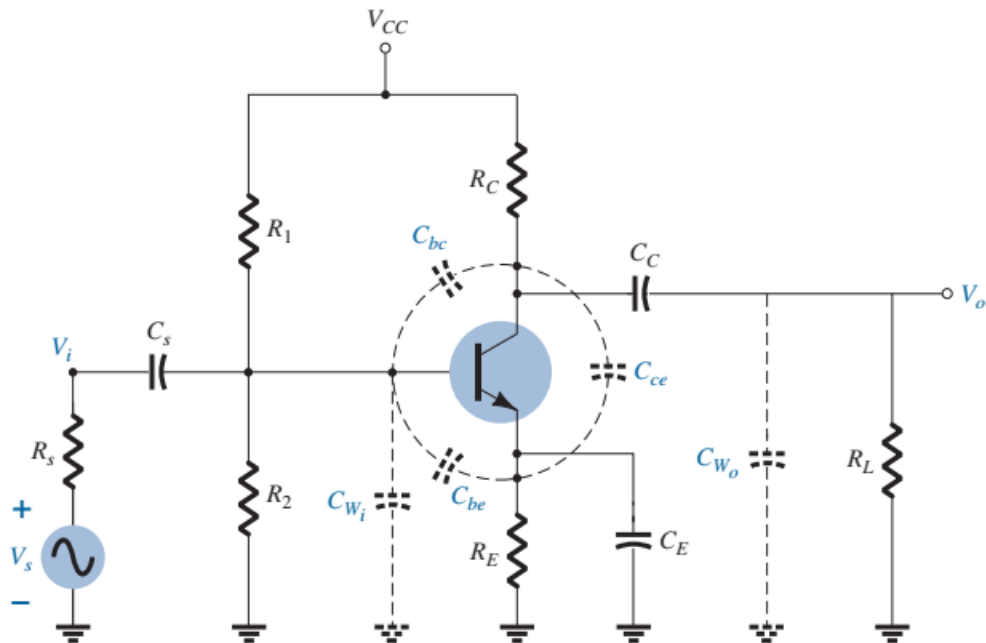
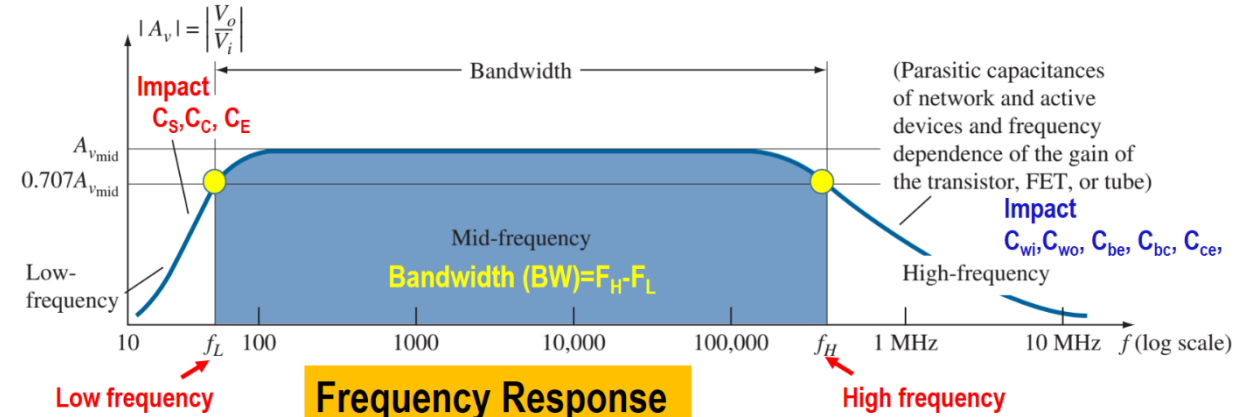
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High-Frequency Response —BJT



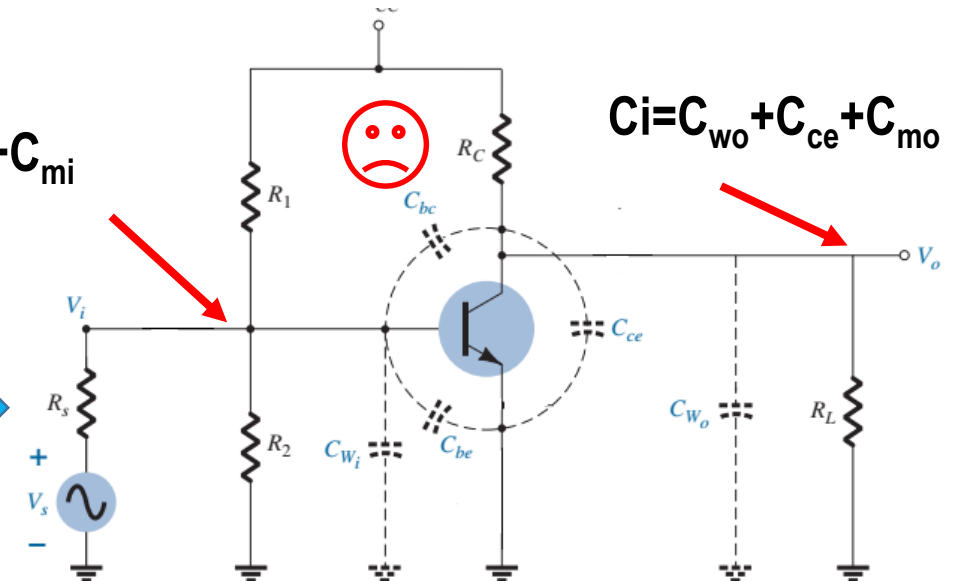
□ Effect of parasitic, wiring, and soldering capacitances

The **large capacitors** of the network that controlled the low frequency response are all replaced by their **short-circuit** equivalent due to their very low reactance levels.



Coupling Capacitors are short circuit

$$C_i = C_{wi} + C_{be} + C_{mi}$$



$$C_o = C_{wo} + C_{ce} + C_{mo}$$

C_{mi} : input miller capacitor

C_{mo} : output miller capacitor

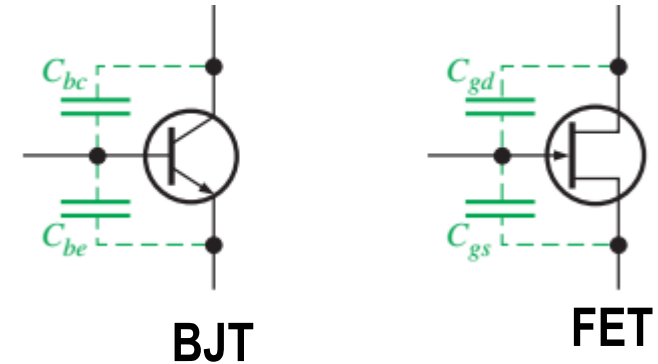


- ☐ Typical Frequency Response
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- ☐ High-Frequency Response —BJT
- ☐ **Miller Theory**
- ☐ Exercise
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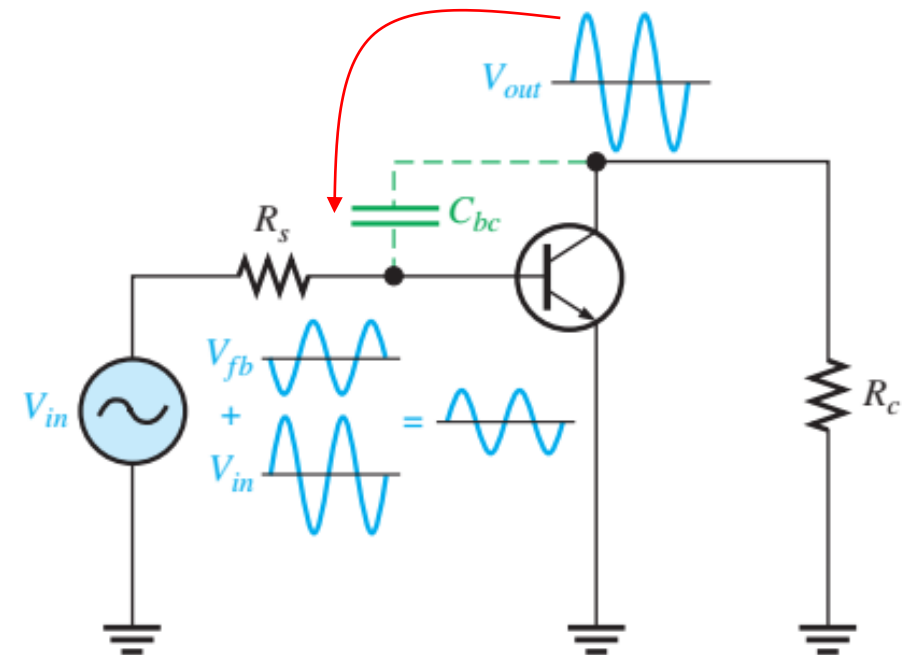
Miller Theory (1/4)



- ❑ Miller's theorem is used to simplify the analysis of inverting amplifiers at high frequencies where the **internal transistor capacitances** are important.
- ❑ Ex, The capacitance C_{bc} in BJTs (C_{gd} in FETs) between the input (base or gate) and the output (collector or drain) is shown in Figure .

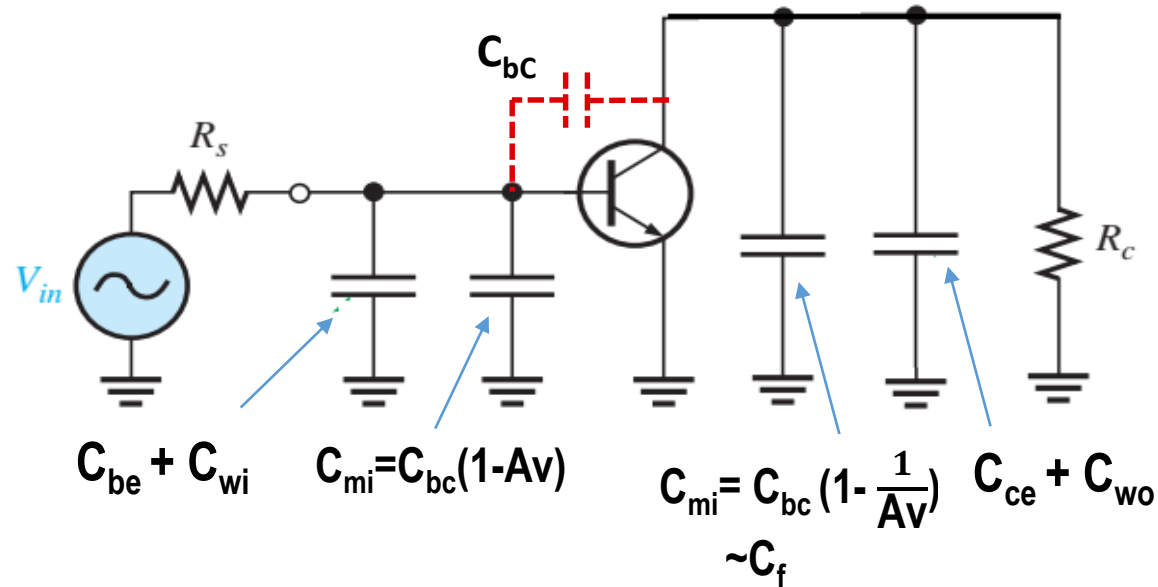


- ❑ Effect of C_{bc} , where part of V_{out} (V_{fb}) goes back through C_{bc} to the base and reduces the input signal because it is approximately 180° out of phase with V_{in} .



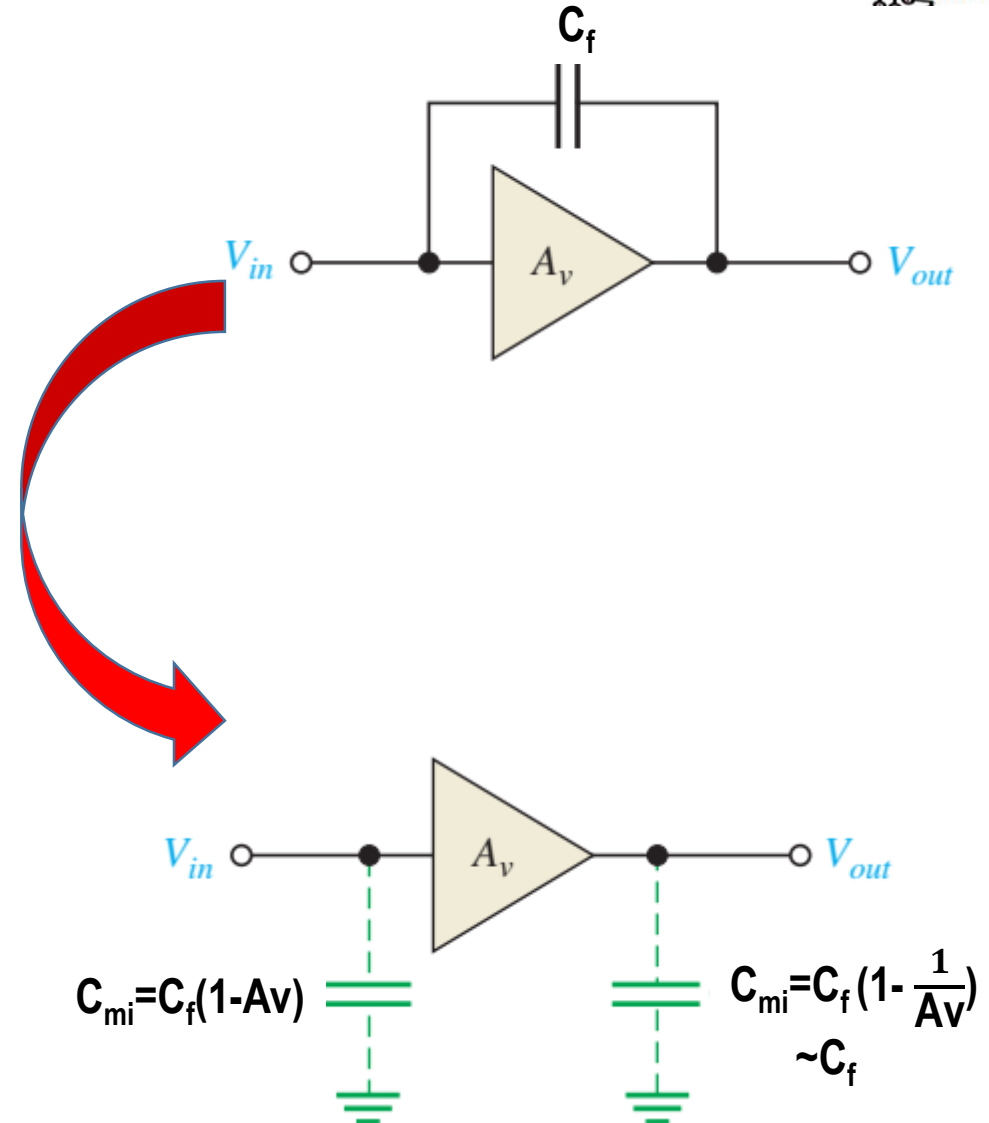
Miller Theory (2/4)

- For any **inverting amplifier**, the input capacitance will be **increased** by a Miller effect capacitance
- For **non-inverting amplifiers** such as the common-base and emitter-follower configurations, the Miller effect capacitance is **not a contributing** concern for high-frequency applications.



$$C_i = C_{wi} + C_{be} + C_{bc}(1 - A_v)$$

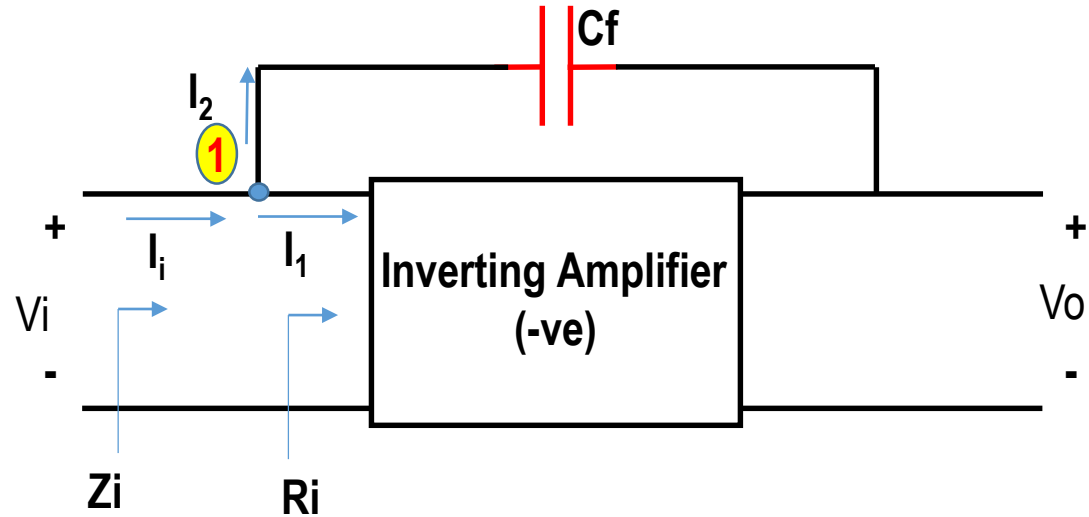
$$C_o = C_{wo} + C_{ce} + C_{bc}(1 - \frac{1}{A_v})$$



Miller Theory (3/4)



□ Prove $C_{mi} = C_{bc}(1-A_v)$ and $C_{mo} = C_{bc}(1-\frac{1}{A_v})$



Node (1)

$$I_i = I_2 + I_1$$

$$\frac{V_i}{Z_i} = \frac{V_i}{R_i} + \frac{V_i - V_o}{X_{cf}}$$

$$V_o = A_v * V_i$$

$$\frac{V_i}{Z_i} = \frac{V_i}{R_i} + \frac{V_i - A_v * V_i}{X_{cf}}$$

$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1 - A_v}{X_{cf}}$$

$$X_{cf} = \frac{1}{sC_f}$$

$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1}{\frac{1}{sC_f(1-A_v)}}$$



$$\frac{1}{Z_i} = \frac{1}{R_i} + \frac{1}{s C_{mi}}$$

$$C_{mi} = C_f (1 - A_v)$$

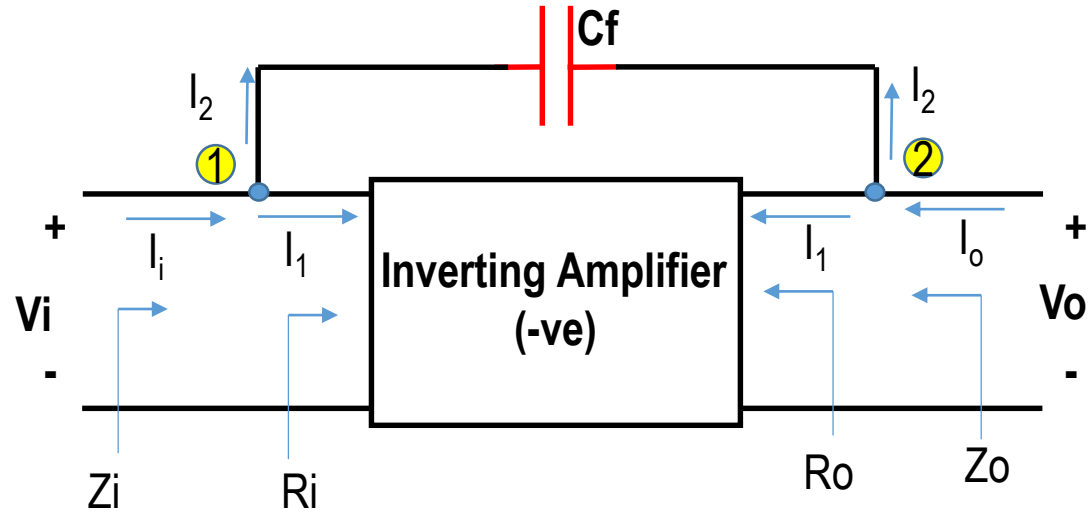


Inverting Amplifier

Miller Theory (4/4)



□ Prove $C_{mi} = C_{bc}(1 - A_v)$ and $C_{mo} = C_{bc}(1 - \frac{1}{A_v})$



Node (2)

$$I_o = I_2 + I_1$$

$$\frac{V_o}{Z_o} = \frac{V_o}{R_o} + \frac{V_o - V_i}{X_{cf}}$$

$$V_i = V_o / A_v$$

$$\frac{V_o}{Z_o} = \frac{V_o}{R_o} + \frac{V_o - \frac{V_o}{A_v}}{X_{cf}}$$

$$\frac{1}{Z_o} = \frac{1}{R_o} + \frac{1 - \frac{1}{A_v}}{X_{cf}}$$

$$X_{cf} = \frac{1}{sC_f}$$

$$\frac{1}{Z_o} = \frac{1}{R_o} + \frac{1 - \frac{1}{A_v}}{\frac{1}{sC_f}}$$

$$\frac{1}{Z_o} = \frac{1}{R_o} + \frac{1}{sC_f(1 - \frac{1}{A_v})}$$



$$C_{mo} = C_f(1 - \frac{1}{A_v})$$



Inverting Amplifier



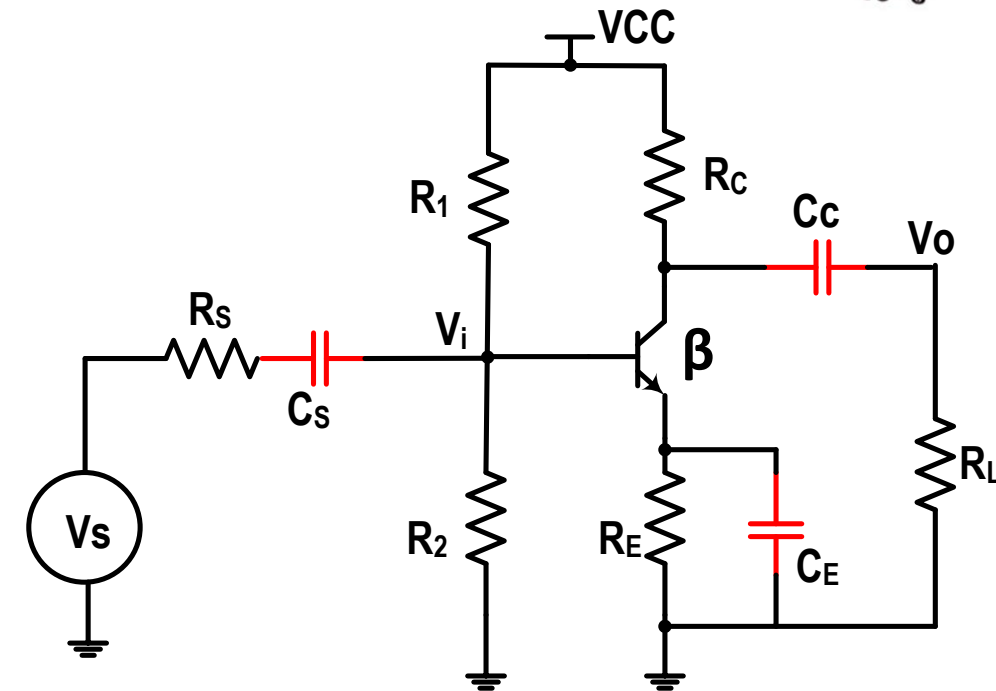
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- ☐ Notes

Exercise (1/4)



□ For the shown BJT amplifier, Determine

1. r_e
2. $A_{v_{(mid)}} = \frac{V_o}{V_i}$
3. $A_{v_{s_{(mid)}}} = \frac{V_o}{V_s}$
4. F_{Hi} and F_{Ho}
5. High cut-off frequency
6. Sketch the asymptotes of the bode plot
7. Sketch the high-frequency response of the amplifier



$$V_{CC}=14\text{ V} , \beta=120$$

$$R_S=0.86\text{k}\Omega , R_1=68\text{k}\Omega , R_2=10\text{k}\Omega$$

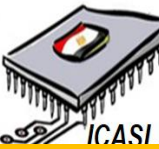
$$R_C=5.5\text{k}\Omega , R_E=1.2\text{k}\Omega , R_L=3.3\text{k}\Omega$$

$$C_S=0.47\mu\text{F} , C_C=0.47\mu\text{F} , C_E=20\mu\text{F}$$

$$C_{bc}=12\text{ pf} , C_{be}=40\text{ pf} , C_{ce}=8\text{ pf}$$

$$C_{wi}=5\text{ pf} , C_{wo}=8\text{ pf}$$

Exercise (2/4)



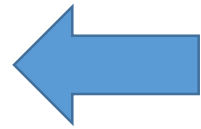
Ans

□ From previous problem

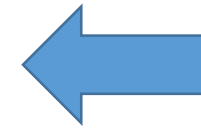
1- $r_e = \frac{26mV}{IE} = \frac{26mV}{1m} = 26 \Omega$

2- $A_{v(mid)} = \frac{V_o}{V_i} = -79.3$

3- $A_{vs(mid)} = \frac{V_o}{V_s} = -57.64$



From DC-Analysis

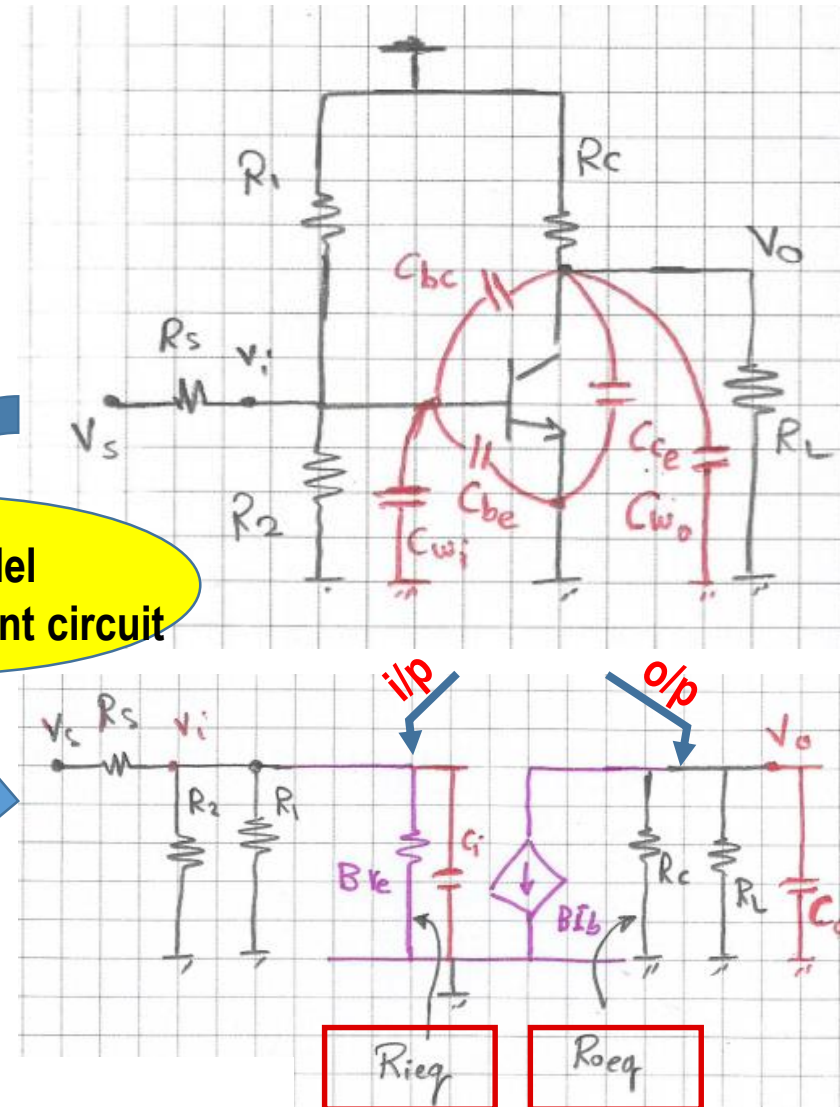


From AC-Analysis

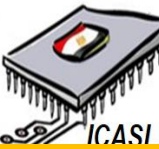
4-To get F_{Hi} and F_{Ho}

- @High frequency, all C_S , C_E , and C_C are ignored. However, the parasitic/wiring capacitances are dominant.
- All dc voltage sources $\rightarrow$ short circuit (Gnd)
- Draw the re-model and AC equivalent circuit
- Get C_i and C_o
- Get R_{ieq} and R_{oeq}

Draw the re-model and AC equivalent circuit



Exercise (3/4)



Ans

$$C_i = C_{wi} + C_{be} + (1 - A_{v(mid)}) * C_{bc} = 5 + 40 + (1 - (-79.3)) * 12 = 1008.6 \text{ pf}$$

$$C_o = C_{wo} + C_{ce} + (1 - \frac{1}{A_{v(mid)}}) * C_{bc} = 8 + 8 + (1 - \frac{1}{-79.3}) * 12 = 28.1 \text{ pf}$$

$$R_{ieq} = \beta * r_e \parallel R_1 \parallel R_2 \parallel R_s = 625.7 \Omega$$

$$R_{oeq} = R_C \parallel R_L = 2.06 \text{ k}\Omega$$

$$F_{Hi} = \frac{1}{2\pi R_{ieq} C_i} = 252.19 \text{ KHz}$$

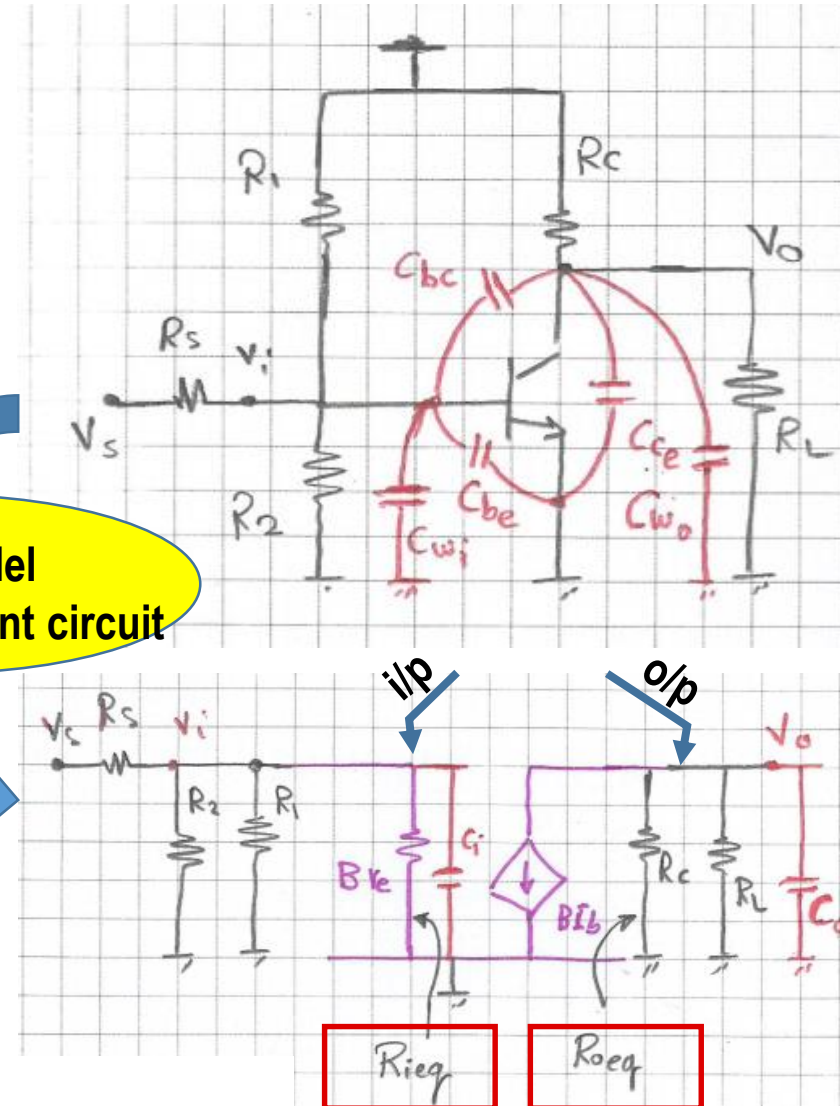
$$F_{Ho} = \frac{1}{2\pi R_{oeq} C_o} = 2.74 \text{ MHz}$$

5-To High Frequency cut-off frequency

$$F_H = F_{Hi} = 252.19 \text{ KHz}$$

The lowest corner frequencies F_{Hi} and F_{Ho}

Draw the re-model and AC equivalent circuit



Exercise (4/4)



Ans

$$C_i = C_{wi} + C_{be} + (1 - A_{v(mid)}) * C_{bc} = 5 + 40 + (1 - (-79.3)) * 12 = 1008.6 \text{ pf}$$

$$C_o = C_{wo} + C_{ce} + (1 - \frac{1}{A_{v(mid)}}) * C_{bc} = 8 + 8 + (1 - \frac{1}{-79.3}) * 12 = 28.1 \text{ pf}$$

$$R_{ieq} = \beta * r_e \parallel R_1 \parallel R_2 \parallel R_S = 625.7 \Omega$$

$$R_{oeq} = R_C \parallel R_L = 2.06 \text{ k}\Omega$$

$$F_{Hi} = \frac{1}{2\pi R_{ieq} C_i} = 252.19 \text{ KHz}$$

$$F_{Ho} = \frac{1}{2\pi R_{oeq} C_o} = 2.74 \text{ MHz}$$

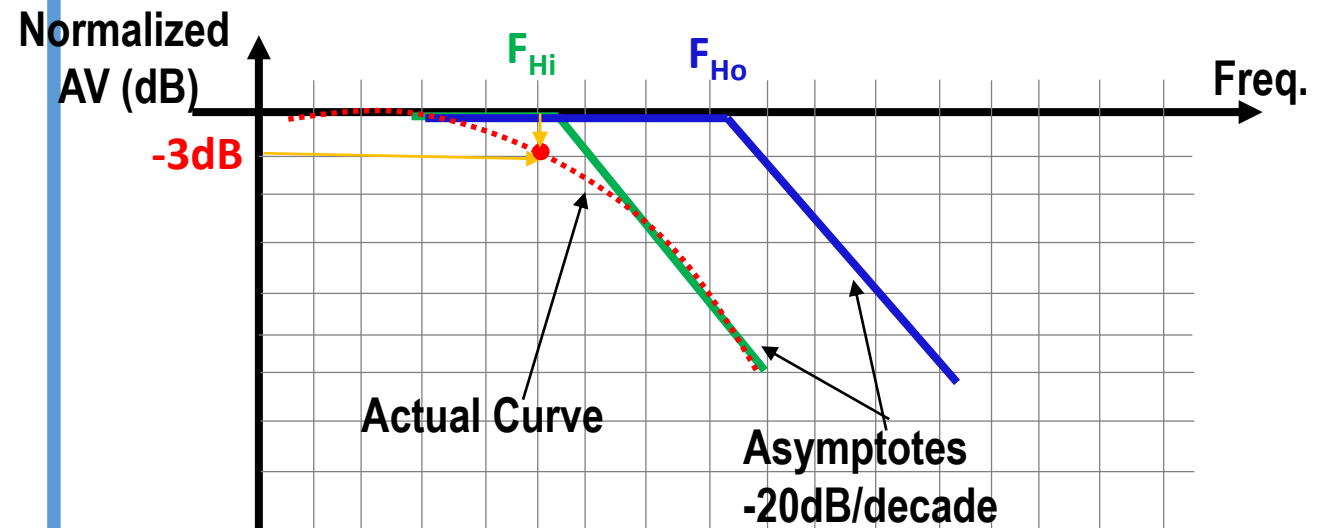
5-To High Frequency cut-off frequency

$$F_H = F_{Hi} = 252.19 \text{ KHz}$$

The lowest corner frequencies F_{Hi} and F_{Ho}



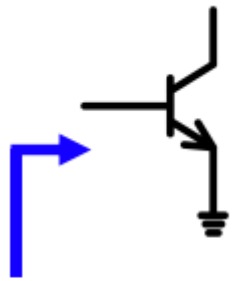
6 & 7-To sketch the asymptotes of the bode plot and high-frequency response of the amplifier





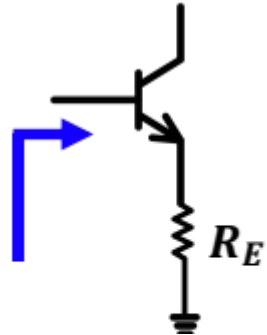
- ☐ Typical Frequency Response
- ☐ Low-Frequency response —BJT
- ☐ Exercise
- ☐ High-Frequency Response —BJT
- ☐ Miller Theory
- ☐ Exercise
- ☐ Notes

Notes (1/2)



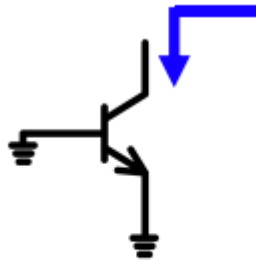
$$Z = r_{\pi}$$

$$Z = \beta r_e$$

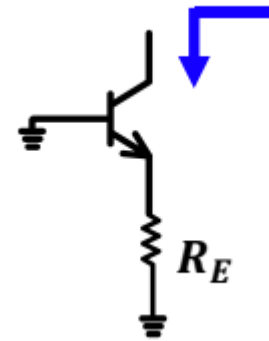


$$Z = r_{\pi} + \beta R_E$$

$$Z = \beta r_e + \beta R_E$$

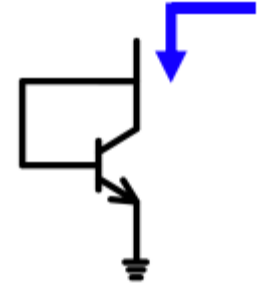


$$Z = r_o$$



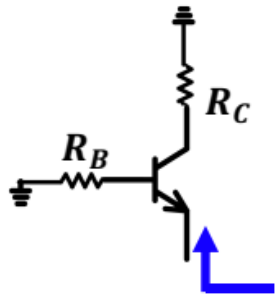
$$Z \cong r_o(1 + g_m(R_E || r_{\pi}))$$

$$Z \cong r_o(1 + \frac{(R_E || \beta r_e)}{r_e})$$



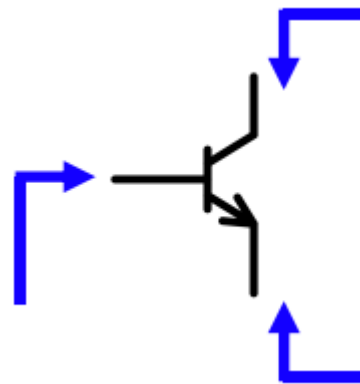
$$Z \cong \frac{1}{g_m}$$

$$Z \cong r_e$$



$$Z \cong \frac{1}{g_m} (1 + \frac{R_B}{r_{\pi}})$$

$$Z \cong \frac{\beta r_e + R_B}{\beta}$$



High impedance



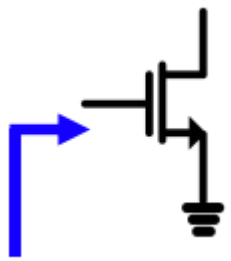
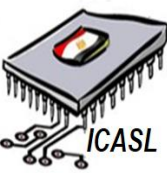
Pole at low frequency

Low impedance

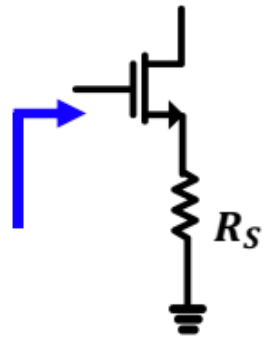


Pole at High frequency

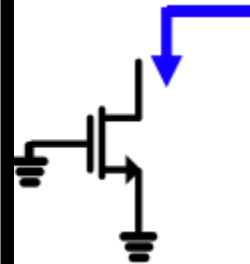
Notes (2/2)



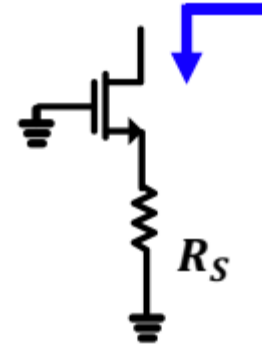
$$Z = \infty$$



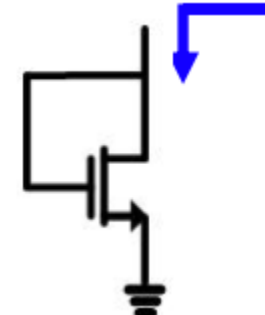
$$Z = \infty$$



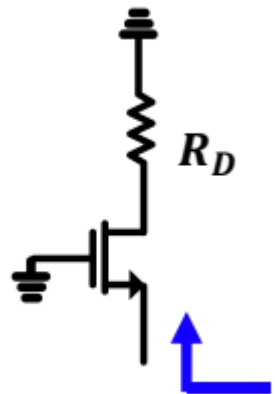
$$Z = r_o$$



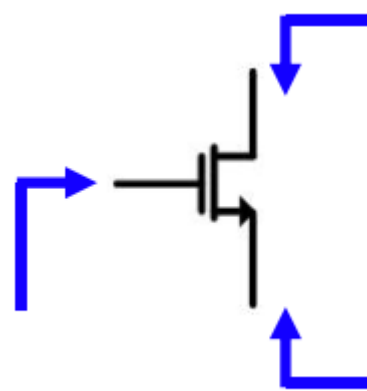
$$Z \cong r_o(1 + g_m(R_E))$$



$$Z \cong \frac{1}{g_m}$$



$$Z \cong \frac{1}{g_m} \left(1 + \frac{R_D}{r_o}\right)$$



High impedance



Pole at low frequency

Low impedance



Pole at High frequency

Thanks!